Source: https://www.fullstackacademy.com/blog/benefits-of-inclusive-design

CSE472: HCI – Human Computer Interaction

Lecture – 14:

Designing for Disabled People

Spring 2026

BRAC UNIVERSITY

Inspiring Excellence

Today

Technology and disabilities

- Who is affected
- Models of disability
- Categories of impairments
- Universal design

Who is affected?

- US Data:
  - 16% of US population to ages 15 to 64 is disabled.
  - 10% of the workforce is disabled
  - 5% of the STEM workforce is disabled
  - 1% of PhDs in STEM are disabled

Metric	Bangladesh	Asia-Pacific Region
Total Population (2025 est.)	175 million	~4.4 billion
Total Persons with Disabilities	2.8% (~4.9 million NSPD 2021) 1.43% (2.36 million Census 2022)	700+ million (16% of population, 2021)
Working-Age (15-64) Prevalence	8-14% functional difficulties	Varies; ~15-16% overall, higher with age

Who is affected?

- People with disabilities
  - Visual, hearing, motor, cognitive, reading
  - About 1 in 4 adults (webaim.org/intro)
- Older adults
  - Up to 50% of computer users may benefit from accessibility features (http://www.microsoft.com/enable/research/)
- "Situational impairments"
  - mobile device users, temporarily injured people
- Sometimes it's just convenient
  - reading transcripts vs. watching a video

Models of Disability

- Medical Model
  - Disabled people are patients who need treatment and/or cure.
- Rehabilitation Model
  - Disabled people need assistive technology for employment and everyday life.
- Legal Model
  - Disabled people are citizens who have rights and responsibilities like other citizens. Accessibility to public buildings and spaces, voting, television, and telephone are some of those rights.
- Social Model
  - Disabled people are part of the diversity of life, not necessarily in need of treatment and cure. They do need access when possible

Models of Disability

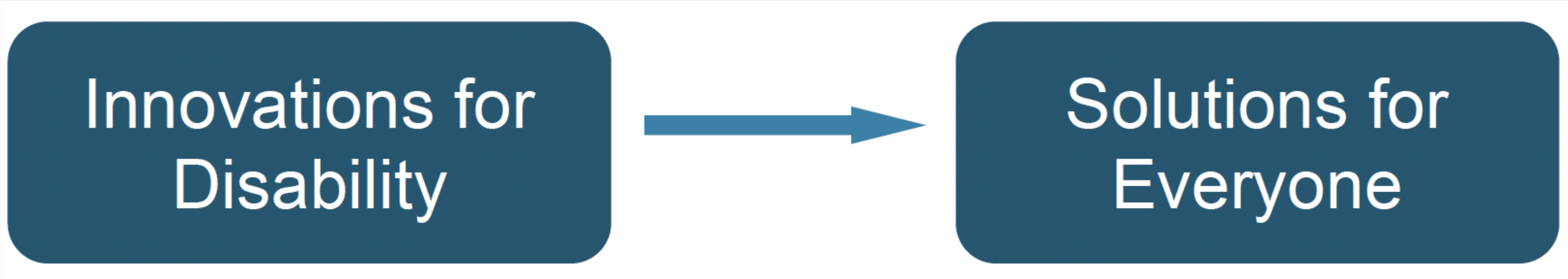
"Disabled people are part of the diversity of life and are not necessarily in need of treatment or cure. They need society to remove barriers and provide access whenever possible."

Technology and Disabilities

- Prosthesis
  - Augmentation to restore lost function. Call it a “cure.”
- Assistive technology
  - Popular in rehabilitation literature.
    - Emphasis on the need for assistance.
- Access technology
  - Allows an activity that would be difficult to impossible to achieve without it.
  - Emphasis not on restoring function, but on achieving an end goal by whatever means possible.
  - Examples: Screen readers, video phones, wheel chairs.

Disabilities Drive Innovation

- Disability and technology innovation are intertwined.
- Information technology fields need more people with disabilities because their expertise and perspectives spark innovation.



- Telephone
- Optical Character Recognition
- Speech synthesis
- Speech recognition
- Texting

- Touchscreens
- Automatic doors
- Video chat
- Closed captions

The Telephone

The telephone was invented by A.G. Bell in his efforts

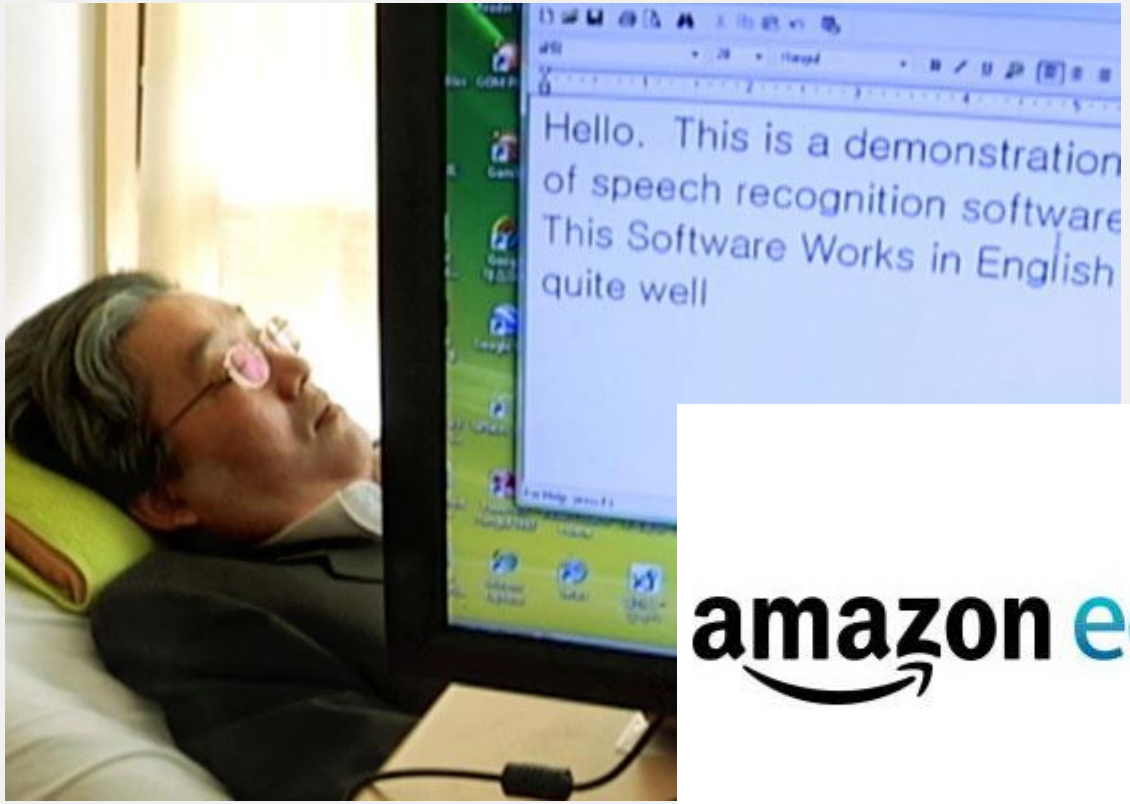
“of devising methods of exhibiting the vibrations of sound optically, for use in teaching the deaf and dumb”

(Fay, American Annals of the Deaf, 1887)

A.G. Bell, 1880

Speech Recognition for Hands Free Access

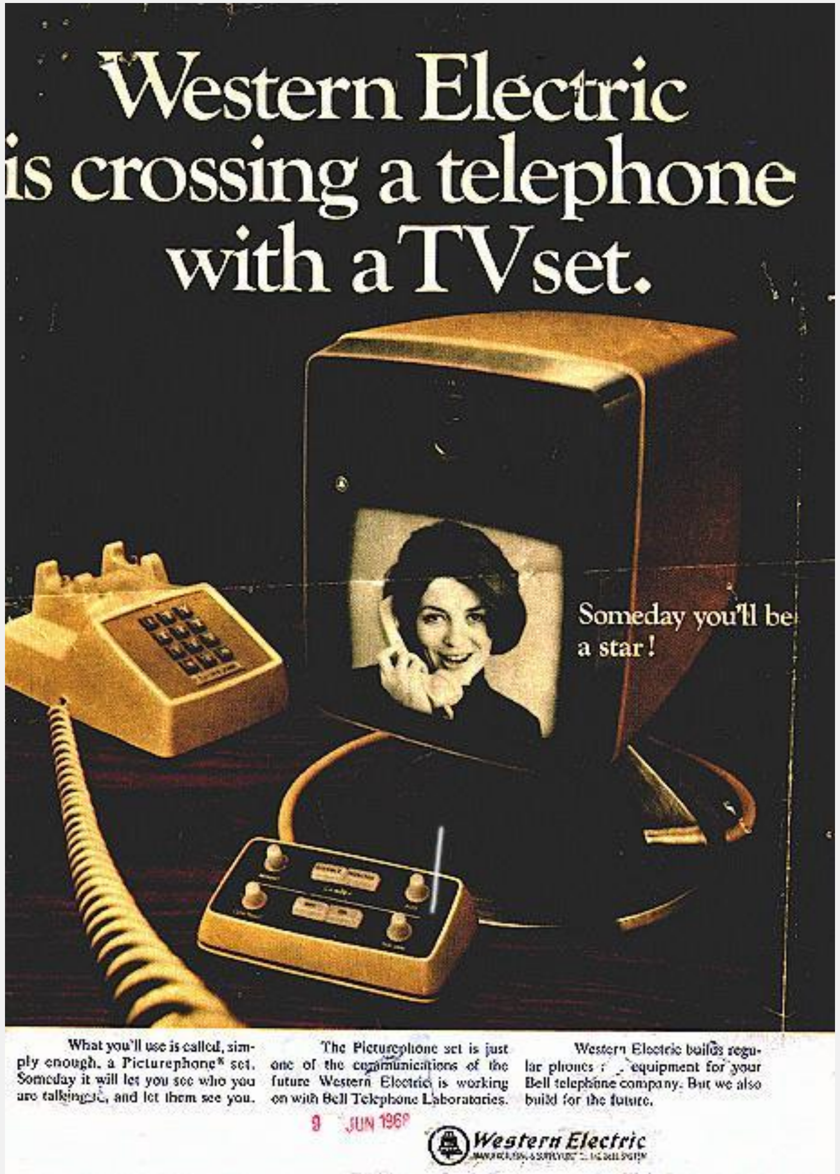
Ray Kurzweil introduced the first commercial large-vocabulary speech recognition software in 1987



Sang-Mook Lee

“Korean Stephen Hawking”

Picturephone

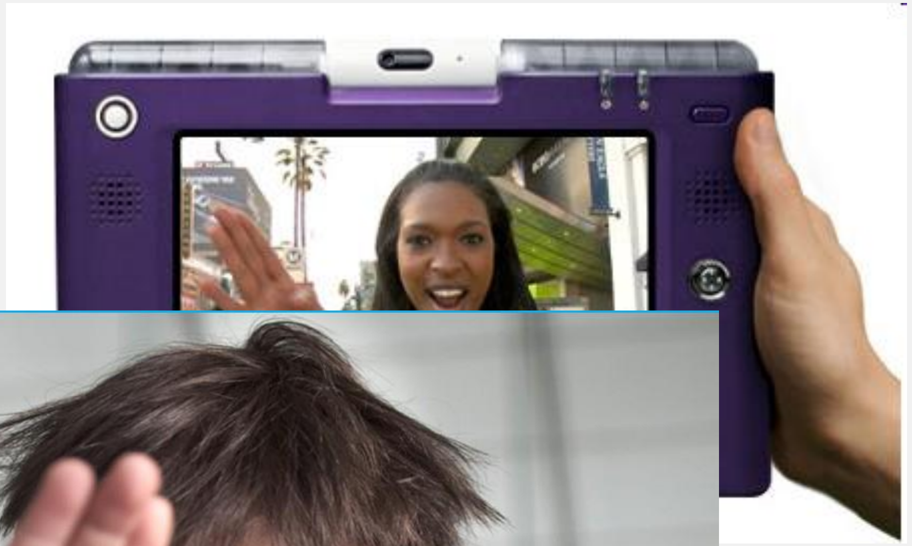


"Picturephone" demonstrated by AT&T at the 1964 World's Fair

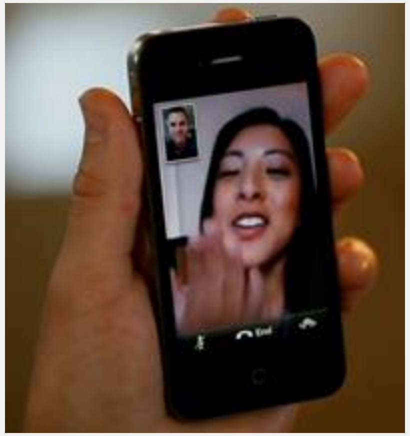
- Required too much bandwidth for phone system
- Deaf World excited then disappointed

Things you will need to consider

Set top box
Sorenson 2002



Purple 2010



Accessibility is Becoming Mainstream

- Accessibility concerns lead to major innovations
- Accessibility is built into products and services
- Companies are focusing on accessibility
    - Microsoft Chief Accessibility Officer
    - Teach Access
- Accessibility is becoming part of the curriculum

Categories of Impairments

- Cognitive (learning disabilities, memory, reading)
- Mobility (Physical)
- Hearing
- Speech
- Visual

Cognitive Impairments

- Memory
  - Working memory, short term memory, long term memory
- Reading
  - dyslexia
- Social
  - e.g. autism
- Learning disabilities
  - ADHD, etc.

- Memory storage
  - Don't rely on users remembering large amounts of information
- Distraction / Task Decomposition
  - Consider users who have difficulty focusing
  - Make tasks shorter, simplify designs

Socialization
  - Some children with autism may not be comfortable looking at faces

Challenges: Physical/mobility

- Diverse array of physical disabilities
- Little or no control of hands
- Temporary injury
- Permanent condition

- Keyboard accessibility
    - Users can access and activate everything with only the keyboard
- Speech recognition compatibility
- Provide ample time for tasks
- Provide shortcuts
    - "Skip Navigation" links


Challenges: Hearing

- Obstacles include videos, mp3s, podcasts
- Often not essential to web content
  - Becoming more essential with things like Siri, Echo.
- Closed-captioning, transcripts
- Sign language
- Hearing aids

- Use structure
  - Use headings and subheadings
  - Use bulleted lists
- Write clearly
  - Keep language short, simple, and to the point
  - Write in active voice
  - Avoid jargon and/or provide definitions
- Provide alternatives to audio
  - Text, captions, and/or sign language interpreters
  - TTY-enabled customer service

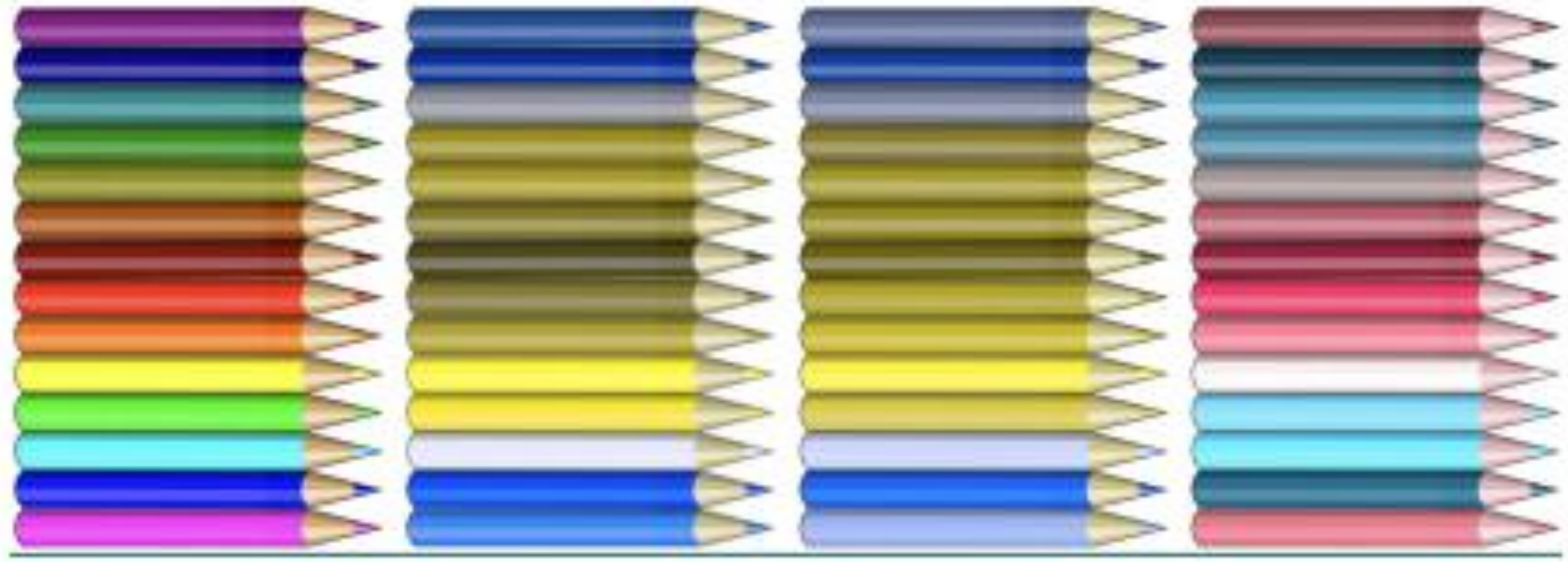
Challenges: Vision

- Many different kinds of vision impairment
  - Blind
  - Low vision
  - Color blind Etc.
- Profoundly affected by web content
  - Web is extremely visual
- Web developers need to accommodate needs more than for any other group

- Use text instead of images of text
  - Use CSS to style text (Logos are exceptions)
- Keyboard accessibility
  - Don't override keystrokes
  - Users can access and activate everything on the page with solely the keyboard
- Skip navigation links
- Have alternatives to color
  - Required fields in red
  - * denotes required fields
- Provide sufficient color contrast

Color Blindless

- Affects 10% of males
- Multiple variations



Simulations of dichromatic color vision. From left to right: original image, simulation of protanopia, simulation of deuteranopia, and simulation of tritanopia. Simulations generated at www.vischeck.org.

Universal Design

- Design for as many users as possible, not just the average user

Seven Principles for Universal Design

Principle 1:

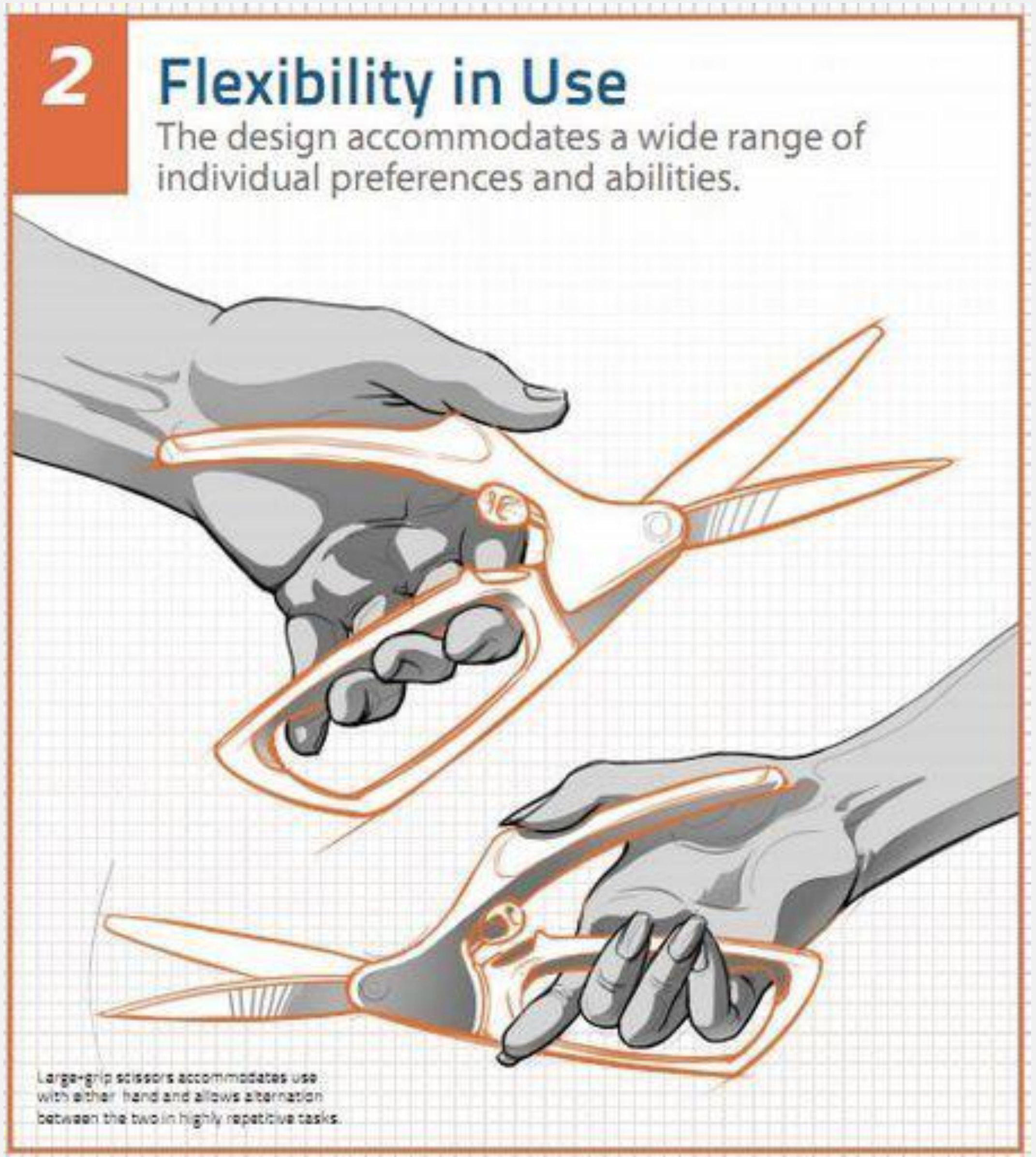
Equitable Use

- The design is useful and marketable to people with diverse abilities.

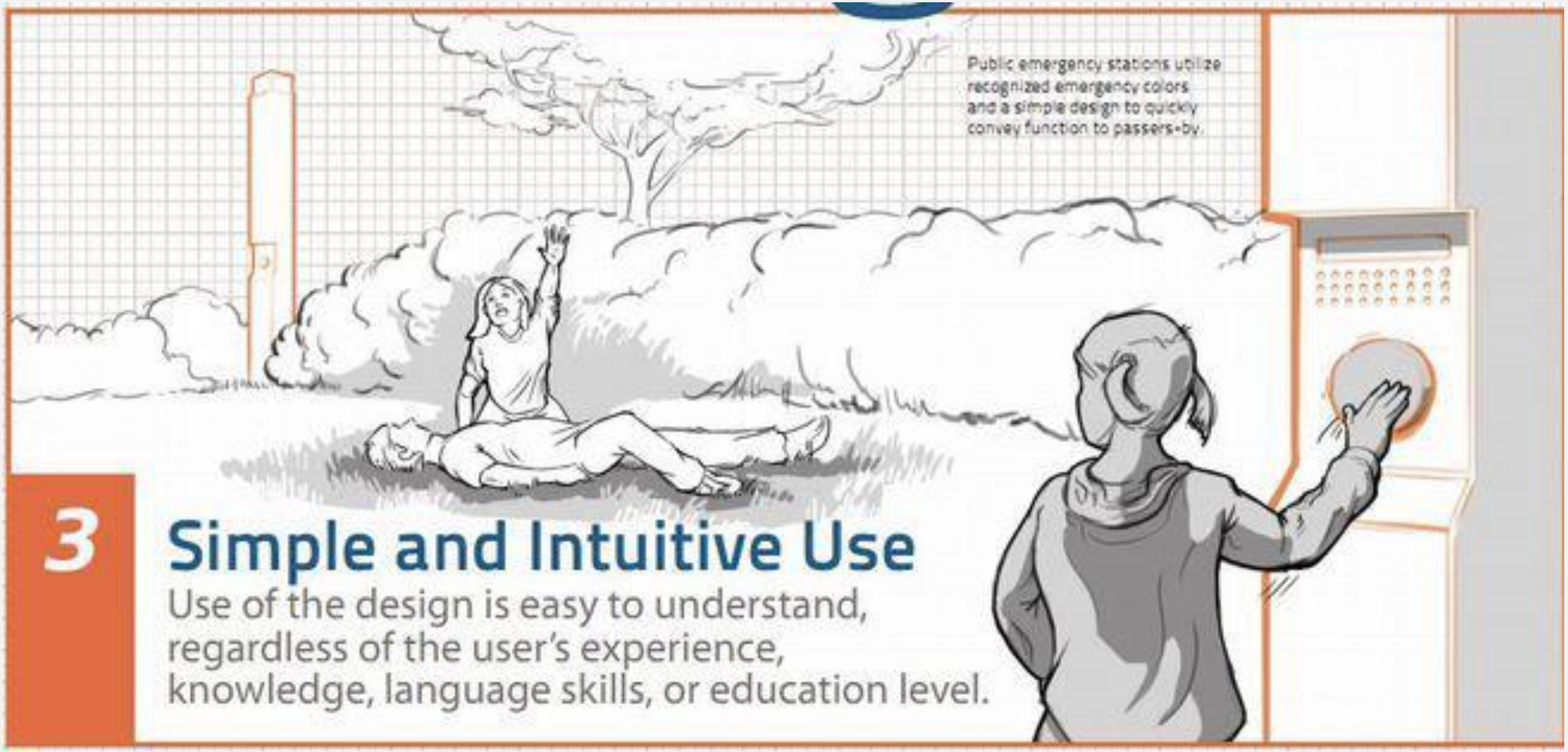
Principle 2: Flexibility in Use

- The design accommodates a wide range of individual preferences and abilities.

- e.g., Many reading apps (iBooks, Kindle) enable users to personalize font size and background color.

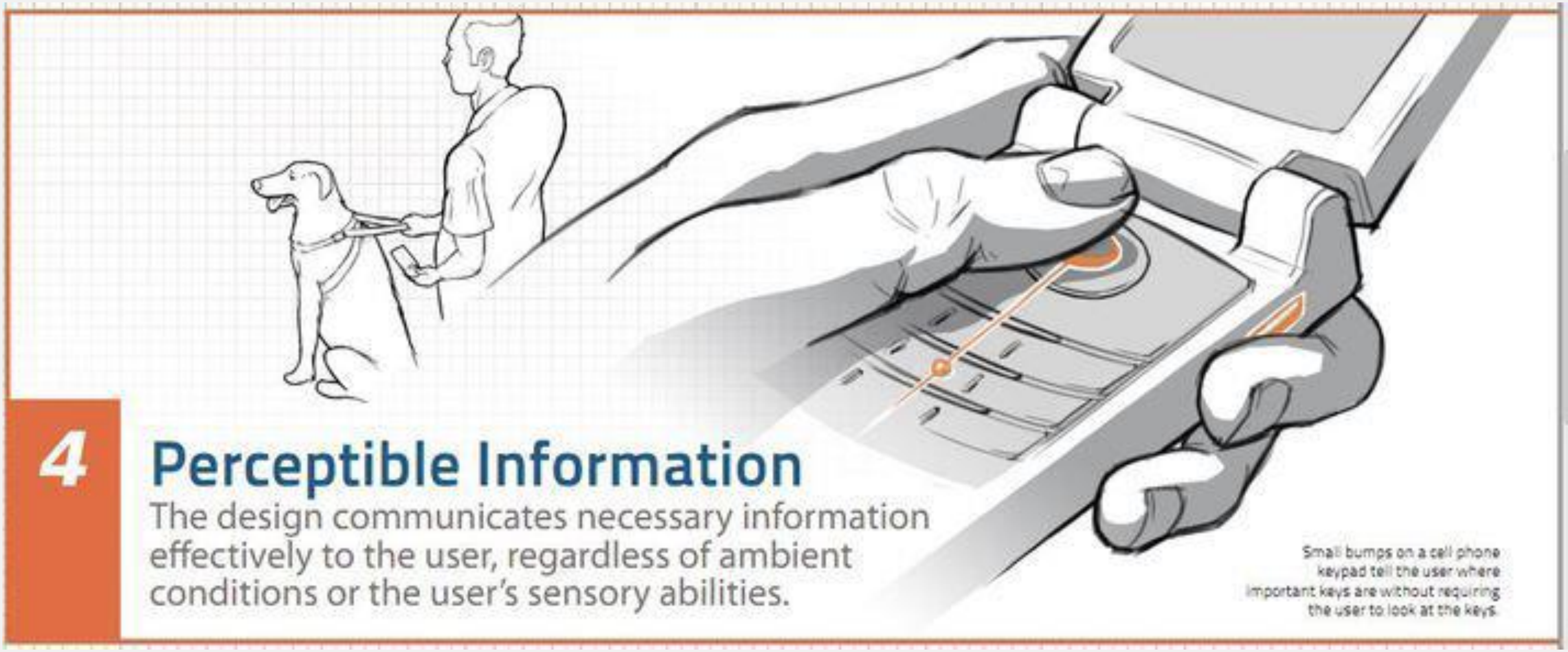


Principle 3: Simple and Intuitive Use



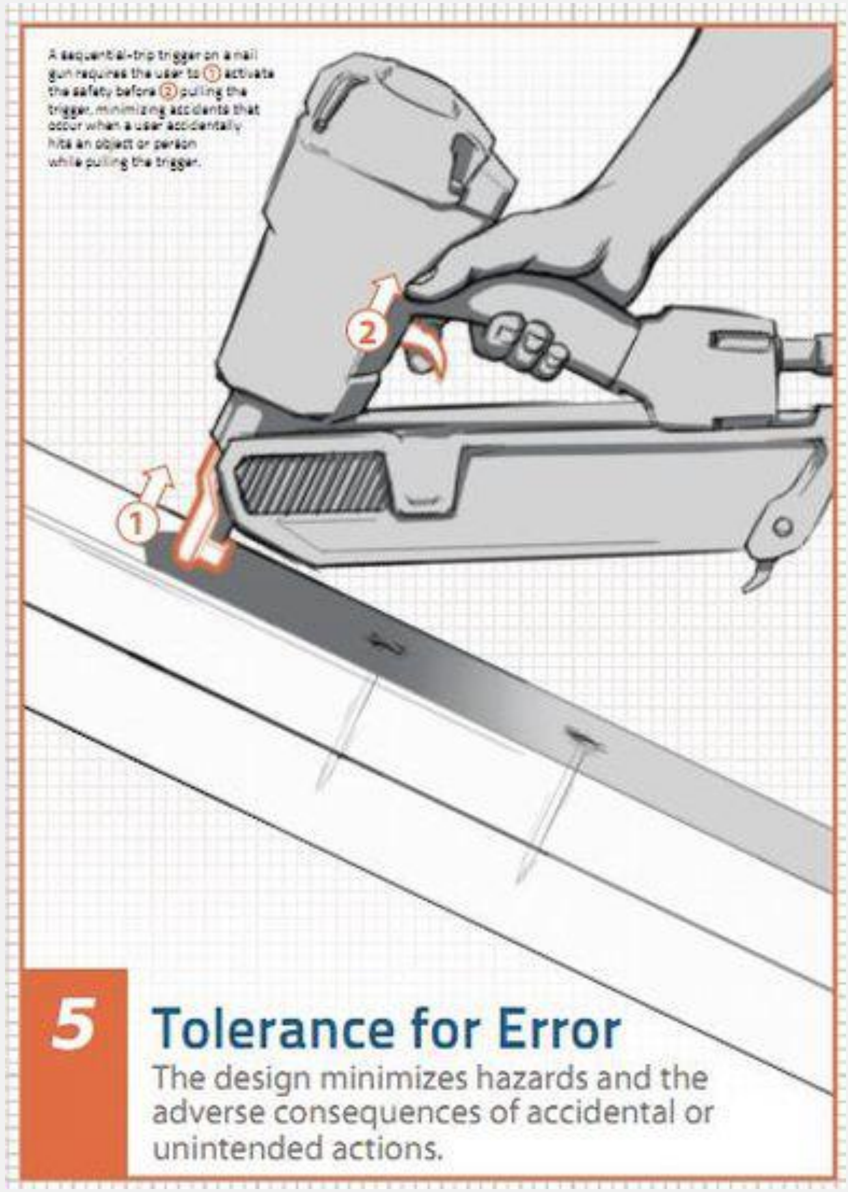
Principle 4: Perceptible Information

- The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.



Principle 5: Tolerance for Error

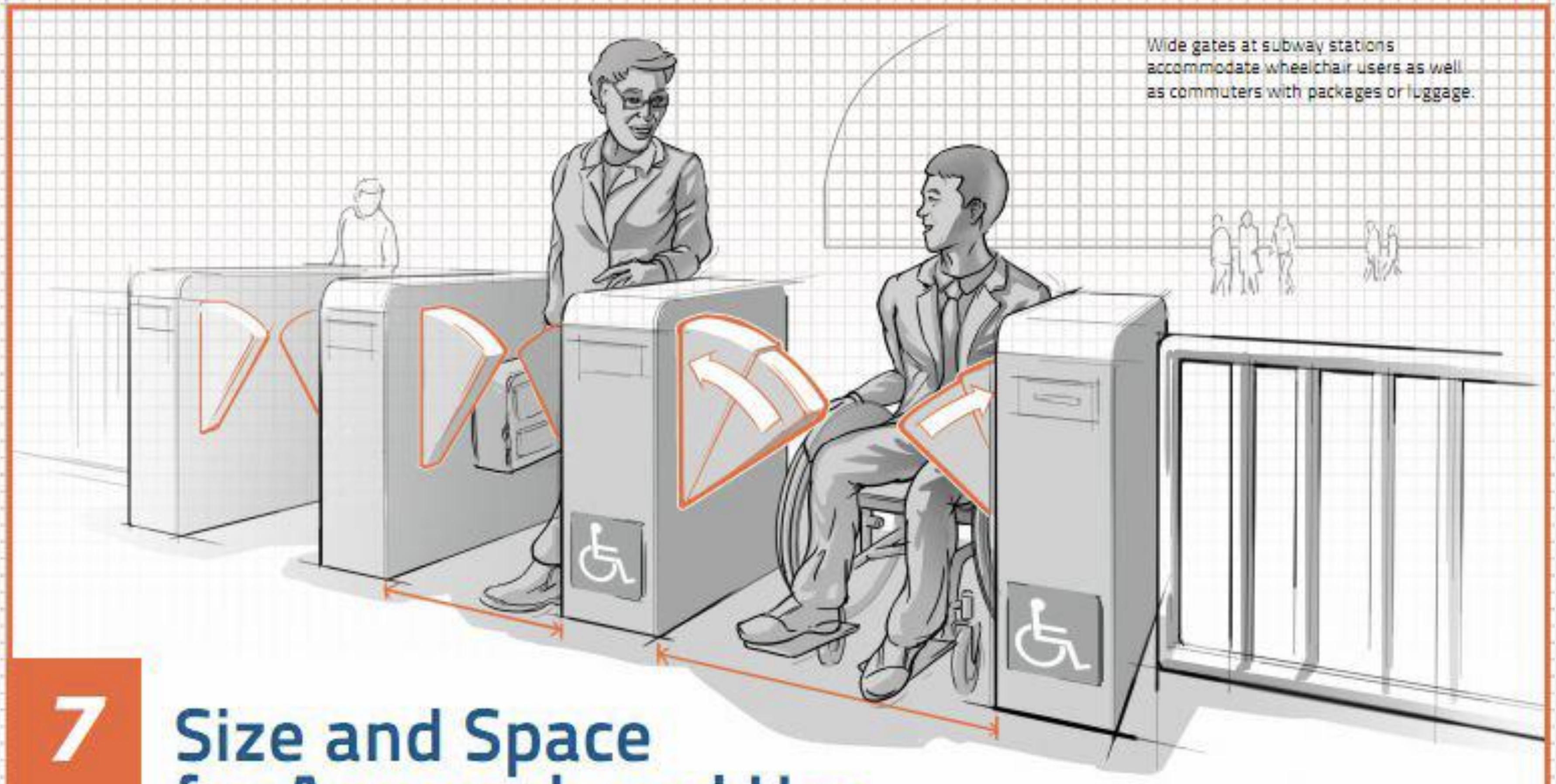
- The design minimizes hazards and the adverse consequences of accidental or unintended actions.
- e.g., Always having an undo or reset button to get back to a familiar state



Principle 6: Low Physical Effort

- The design can be used efficiently and comfortably and with a minimum of fatigue.
- e.g. Adding appropriate amounts of white space decreases the physical cognitive effort.

Principle 7: Size and Space for Approach and Use



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Size and Space for Approach and Use

Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

Summary

- Technology and disabilities
  - About 1 in 5 people affected
  - Many more, depending on how you count…
  - There are different models of disability

- You need to:
  - Understand categories of impairments
  - Design for accessibility
  - Practice Universal design

Any questions?

Acknowledgment: Lecture content adapted from Cornell University HCI & Design (Prof. Nicki Dell)